DSA0308 - NATURAL LANGUAGE PROCESSING

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CIA - 1

① Regular Expression:

(a.) dd/mm/yy date

```
\b(a[1-9]|[12][0-9]|3[01]|(0[1-9]|1[0-2])/
    [0-9]{4}\b
```

(b.) mobile number

```
(?<!\d)(?:(?:\+91[-]?|0)?[6-9]\d{9}
    (?!\d)
```

c.) email address

```
\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]
    +\.[A-Za-z]{2,}\b
```

Line	Date	Mobile No.	Email
L_1	05/03/2025	None	None
L_2	None	9876543210 +91-9123456789	None
L_3	None	None	None
L_4	None	None	ravi.k@college.edu.in

Mobile number pattern without anchoring or look-around produces a false positive in L3

" [6-9]\d{9} " → This will match first 10 digits of the mobile number.

Corrected pattern: (?<!\d)([6-9]\d{9})(?!\d)

This ensures the mobile number is not part of long sequence.

Date substitution:-

Replace with $3-$2-$1

Example:

05/03/2025

↓

2025-03-05

② Given lexicon:

Reg. noun = { fox, cat, dog }

irreg-sg-noun = { goose, mouse }

irreg-pl-noun = { geese, mice }

a.) Finite-state automation

q_0	Start state
q_1	Reg. noun
q_2	Irreg-sg-noun
q_3	Irreg-plural-noun
q_f	Final state

Transition:

From	Input	To
q_0	fox, cat, dog	q_1
q_1	-s	q_f
q_0	goose, mouse	q_2
q_0	geese, mice	q_3
q_2	End	q_f
q_3	End	q_f

Reg. noun accept the plural suffix -s

Irreg. noun (plural) accepted a complete words.

b.)

Sequence	Result
dog + -s	✓
mouse + -s	X
geese	✓
fox + -s	✓
mice + -s	X

c.) Different levels.

Word	Lexical Level	Intermediate level	Surface level
foxes	fox + s	fox + es	foxes
cats	cat + s	cat + s	cats
Geese	goose + PL	geese	geese

When the noun ends in

x, s, sh, ch (or) z

the plural is formed by adding -es

for Example:

fox + s → fox + es → foxes

Inflectional Vs derivational Morphology

Word	Types	Justification
Walked	Inflectional	Past tense of walk
unhappiness	Derivational	create new word from happy
computerization	Derivational	Create new word from computerize
Geese	Inflectional	Irreg. plural of goose.

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(3). **Porter Stemmer:** It reduces the word by using suffix removal rules.

Word Given :-

Ponies, monitoring, relational, hopefulness and generalizations.

word	Step	Rule	Result
Ponies	1a	ies → i	Poni
Monitoring	1b	ing → removed	Monitor
Relational	2	ational → ate	Relate
hopefulness	3	fulness → ful	hopeful
generalizations	1a	s → removed	generalization

Generalizations and general both reduce to the same stem. It is because

Generalization undergoes suffix removal through the porter stemmer algorithm until it becomes general.

Therefore, words of both reduce to the same stem ie, general.

(ii.) Given:

<s> the cat sat </s>

<s> the cat ran </s>

<s> the dog sat </s>

unigram:-

Word	count
the	2
cat	1
dog	1
sat	1
ran	1
<s>	3

Total predicted token = 12

Bigram count

Word	count
</s>	2
ran	1

Bigram Probabilities formula :-

$$P\left(\frac{w_i}{w_{i-1}}\right) = \frac{C(w_{i-1}, w_i)}{C_{w_{i-1}}}$$

S_1 = the cat sat

$$P(S_1) = \frac{3}{3} \times \frac{2}{3} \times \frac{1}{2} \times \frac{2}{2}$$

$$= 1/3$$

S_2 = the cat ran

$$P(S_2) = \frac{3}{3} \times \frac{1}{3} \times \frac{1}{1} \times \frac{2}{2}$$

$$= 1/3$$

S_3 = the dog sat

$$P(S_3) = \frac{3}{3} \times \frac{1}{3} \times 0 \times \frac{1}{1}$$

$$= 0$$

Cross - Entropy for S_1

$$H = -\frac{1}{4} \log_2 P(S_1)$$

$$= -\frac{1}{4} \log_2 (1/3)$$

$H \simeq 0.3963$ bits / word

④ a.)

unigram	
<s>	3
I	3
like	2
tea	2
coffee	1
drink	1
</s>	3

Bigram	
(<s> I)	3
(I like)	2
(I drink)	1
(like tea)	1
(like coffee)	1
(drink tea)	1
tea </s>	2
coffee </s>	1

$$P(\text{Tea} \mid \text{like}) = \frac{1}{2} = 0.5$$

$$P(\text{drink} \mid \text{like}) = \frac{0}{2} = 0$$

unsmoothed Probabilities,

$P(\text{Tea} \mid \text{like}) = 0.5$ & $P(\text{drink} \mid \text{like}) = 0$

b.) $$P(w/v) = \frac{c(v,w)+1}{c(v)+V}$$

$$P(\text{tea} \mid \text{like}) = \frac{1+1}{2+6} = \frac{2}{8} = 0.25$$

$$P(\text{drink} \mid \text{like}) = \frac{0+1}{2+6} = 0.125$$

Reconstituted count:

$$c^* = p(\text{tea}|\text{like}) \times c(\text{like})$$

$$= 0.25 \times 2$$

$$c^* = 0.5$$

c) backoff with $d = 0.5$

$$p^*(\text{Tea}|\text{like}) = p^*(\text{coffee}|\text{like})$$

$$= \frac{1-0.5}{2} = 0.25$$

Now $p^*(\text{coffee}|\text{like}) = 0.25$

Left-over mass $\beta(\text{like}) = 1 - (p^*(\text{tea}|\text{like}) + p^*(\text{coffee}|\text{like})$

$$= 1-(0.25+0.25)$$

$$= 0.5$$

$$p(\text{tea}) = \frac{2}{12} \quad ; \quad p(\text{coffee}) = \frac{1}{12}$$

Total:

$$1 - \frac{2}{12} - \frac{1}{12} = \frac{9}{12} = 0.75$$

$$\alpha(\text{like}) = \frac{0.5}{0.75} = \frac{2}{3}$$

$$P(\text{drink} \mid \text{like}) = \alpha(\text{like})\, P(\text{drink})$$

$$= \frac{\alpha'}{3} \times \frac{1}{6}$$

$$= \frac{1}{18}$$

d.) Deleted Interpolation :-

Given: $\lambda_2 = 0.7$, $\lambda_1 = 0.3$

$$P(w/v) = \lambda_2 P_{\text{bigram}}(w/v) + \lambda_1 P_{\text{unigram}}(w)$$

$$P(\text{drink} \mid \text{like}) = 0.7(0) + 0.3\left(\frac{1}{12}\right)$$

$$= 0 + 0.025$$

$$= 0.025$$

$$P(\text{tea} \mid \text{like}) = 0.7(0.5) + 0.3\left(\frac{2}{12}\right)$$

$$= 0.40$$

e.) Comparison & Justification :-

unsmoothed	0.500	0
Adding one	0.250	0.125
Backoff	0.250	0.056
deleted interpolation	0.400	0.025

(5)

a)

V_1 (PRP) $= 0.5 \times 0.8$

$= 0.4$

$V_2 \rightarrow$ "can"

$V_3 \rightarrow$ fish

MD $0.4 \times 0.4 \times 0.3$

$= 0.048$

VB $0.4 \times 0.5 \times 0.02$

$= 0.004$

max $(0.048 \times 0.8, \quad 0.004 \times 0.1, \quad 0.0004 \times 0.4) \times 0.01$

$= 0.000384$

NN $0.4 \times 0.1 \times 0.01$

$= 0.0004$

max $(0.048 \times 0.2, \quad 0.004 \times 0.5 \times 0.0004 \times 0.3) \; 0.02$

$= 0.000192$

Therefore the final state is VB

(b) Rule - based :

Rule 1 :-

"can" → MD, VB, NN

Previous words → personal pronoun.

so it removes the NN & VB

can becomes the MD.

Rule 2 :-

If Previous words /tags → MD

⇒ it removes NN.

∴ fish = VB

(c) Transformation :

They | PRP can | MD fish | NN

all have a one - error out of these (3) tokens.

change NN to VB → when the previous tag is MD

Therefore it changes the 0 error from 1 error

∴ 46 - 6 = 40

(d.) Comparison: Tagging

Rule - Based:

* It is hand - written. so it is high cost
* Transparent and also hard to maintain it.

Stochastic - based:

* uses probabilities from a tagged corpus.
* Sparse - data problems.

Transformation - based

* starts with initial tags and applies ordered rules